9 **(a)** An experiment directs alpha particles at a very thin sheet of gold foil.

(i) Most of the alpha particles pass through the thin foil in a straight line.

State the conclusion about atoms from this observation.

.....

..... [1]

(ii) Some of the alpha particles are deflected through angles less than 90° and a few are deflected through 180°.

State and explain **two** conclusions about the nuclei of atoms from this observation.

conclusion 1 .....

explanation 1 .....

.....

conclusion 2 .....

explanation 2 .....

.....

[4]

(b) A source contains a radioactive isotope of strontium. This isotope decays by emission of β-particles. The half-life of this isotope is 29 years.

(i) State the change in the nucleus which occurs when a β-particle is emitted.

..... [1]

(ii) The initial mass of this isotope of strontium in the source is 25 μg.

Calculate the mass of the strontium isotope that decays in 87 years.

mass = ... μg [3]

[Total: 9]

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